

Improvement of Honey Production: A Smart Honey Bee Health Monitoring System

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Abstract—For commercial purpose, only 2 species are mostly used for farming which is *H. Itama* and *G. Thoracica* that very active in the production of honey and propolis. It is important to keep the beehives in good condition for the bees working. Colony loss is the main problems were affected by many of factors. The stingless bees are very sensitive to the surrounding changes. With an eye toward lessen the loss of colony and honey production impact, a development of Internet of Things (IoT) monitoring stingless bees and stingless bee health is being carried out at stingless beehive to monitor the habitat environmental need for productive and healthy colonies. Therefore, the objective of this project are to investigate the parameters to be monitored that affect the behaviour of the stingless bees and production, to select suitable sensors to measure the parameters, to integrate and test the sensors on a selected microcontroller system, to investigate and integrate Internet of Things (IoT) platform and to demonstrate the prototype for monitoring stingless beehives and stingless bee health. The parameter that will be measured including temperature and humidity inside the hives, smoke gas around the hives and weight of the hives are measured by using suitable sensors. With Wi-fi connection, the data can be transfer to the Cloud and able the beekeepers to monitor on User-Interface at the webpage on any smart devices for the status update. In this preliminary implementation, the IoT real-time monitoring system to measure parameters of temperature, humidity, weight, and smoke of hive has been built. The behavior and health of stingless bees have been analyzed. Therefore, it can be concluded that the implementation of IoT monitoring could solve the problem of colony loss face by the beekeepers.

Keywords—Bee; IoT; health.

I. INTRODUCTION

The honey Bee industry considered as a highly profitable industry in Malaysia that imported RM 50 millions of honey product in 2013 [1]. The Bee products consist of three main different products namely honey, Bee bread and propolis. Potential marketing of stingless Bee product is very positive in both local and overseas markets [2]. To improve the productivity and quality of honey for commercialization, farming methods will carry out proper scientific and

autonomous ways. In other country, they have develops the way they check the beehives and Bee health by wirelessly remote monitoring.

By monitoring the temperature, humidity, smoke and the weight of Bee hive, the bee-keeper can get the information about the queen lay, the activities of colonies, presence of pests and the level of honey produced. It enables bee-keepers to get real-time information from hives. A database allows partners in biology to study pests and diseases. Sensor fitted to the beehives will detect temperature, humidity, smoke gas and the weight and send the real-time result to the beekeepers via Cloud so that they can check and analyze the bee activity through Internet [3]. These weight scales typically take readings often and capture small changes in weight that show various colony activities such as the departure and return of foragers, nectar processing, etc. If the bees have difficulty maintaining temperature, it could mean the queen is not lay, there is not enough nursing bees to heat the hive is starving. If the relative humidity is low, bees are not consuming honey [4].

A scalable and cost-efficient WSN was used to monitor the colonies. However, further studies by expanding the scope of the system could provide more information. A decision support system can be developed using geographical information systems to provide information for the migratory beekeepers about the region they will visit and to find a route for them with the highest honey yield potential. As a result, it will be possible to ensure the optimum use of the nectar sources and to take maximum advantage of the beekeeping activities [6]. According to the authors, they had created an economical, stable beehive-monitoring system. The adapter board contains a simple adaptation circuit of RS232 levels to minimise the power consumption of the node for the undesirable weight data using a 3.7 V/2300 mAh rechargeable lithium-ion battery. More autonomy can be obtained by increasing the acquisition period. By connecting to the local database server, the data of colonies are available on the Internet via unique cloud database server and the Internet [7]. The authors are look forward to increase new sensors in the future to monitor other parameters such as the weight of a beehive, atmosphere and illumination.

As a result, beekeeper and researchers will be able to constantly monitor various environmental conditions of honeybee colonies [8-16].

Electronic bee monitoring technology has grown recently to enable user-friendly data collection unobtrusively. From the works that have been done previously, there is a development for the monitoring beehives method which is similar objective where to investigate the parameters to be monitored that affect the production of honey. Temperature, humidity, smoke gas around the hives and weight of the hives parameters will be investigated. Furthermore, suitable sensors to measure the parameters can be selected where the common sensor will be used such as SHT22 which more accurate range than SHT11 sensor module. The sensors can be integrated and tested on a selected microcontroller system. Microcontroller with built in Wi-Fi connectivity is more recommended where the speed can be improved without external Wi-Fi. Wifi is chosen rather than others IOT technologies such as Zigbee because Wi-Fi is wider typical distance coverage at about 30 to 100 meters and can work up to 54 Mbps of data rate where it is suitable for application of extension of cellular connection within home [9]. Then, the prototype for monitoring stingless beehives and stingless bee health will be demonstrated at two different places to investigate the factors affecting the hives.

II. SYSTEM DESIGN

A. Overview

This section shows an attempt to design research of a monitoring device to monitor the stingless beehives using Internet of Things Application (IoT). With these sophisticated technologies, this project can get the real-time information and data of the stingless beehives from anywhere else around the world. Figure 1 shows a beehives ready for producing honey. The temperature, humidity, smoke gas and weight sensor were installed to the beehives will send data to the user interface. The data is sent with the assist of the Cloud. The data is received with the combination of wireless sensor and Cloud through Wi-Fi so that an output will be appeared on the LCD Display and to the User Interface, MQTT lens. The Cloud will give information about the temperature, humidity, smoke gas and weight so that the user can monitor the condition of the stingless beehives. The optimum parameters that have been studied are temperature of 28- 30 degree Celsius, humidity 65% - 75%, Weight is unknown and Smoke Gas 300 -10,000 ppm.



Fig. 1. Model of a stingless beehives

B. Internet of Things (IoT)

IoT is simply connecting all the surrounding smart devices (things) to the Internet as shown in Figure 2. It represents as nervous system that connects anything with everything on the Earth. Sensors are built in everywhere, routinely collecting real-time data. Each device has its unique ID to allow communication between them of machine to machine (M2M) communication. IOT works with one device with built in sensors and actuators send the data to Internet, then the data will be received by another device to take necessary smart decisions. Cloud is the solution for storing a huge amount of data which provide limited cloud space for free of cost. With one smart network infrastructure, operations can be improved, safety and security will be increased, and valuable data insight can be gained to streamline automation. In today's challenging industrial environments, the decent IoT wireless connectivity is a must.

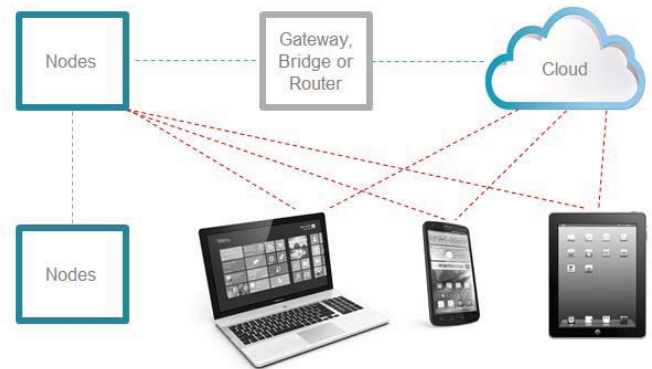


Fig. 2. IOT Working Structure

C. Proposed Design of the System

In this section, all the material and method used in the project will be described. All of the matters about the material and method used in the project including block diagram, the hardware and software used. In most projects, the methods and material is one important element that should be highlighted.

For experimental method to investigate the factors affecting the activities, health of stingless bees and honey production, the device built will be tested on 2 different places

of the hives. The test will be carried out in 2 months. The first experiment will be at the place which is surrounded by the flowers garden or orchards while the second place will be at residential area.

D. Block Diagram

The block diagram is designed to show overview on how the system is designed.

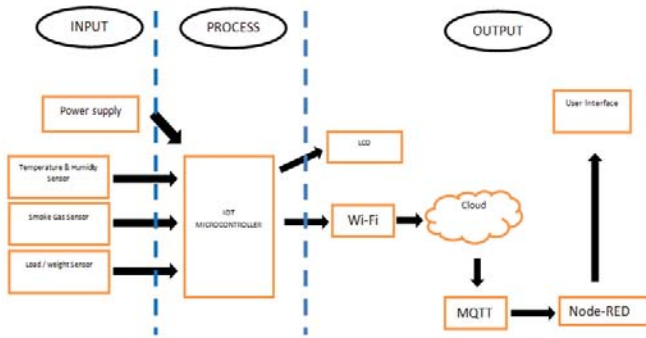


Fig. 3. Block Diagram

The block diagram of the project is divided into 3 parts, input, process and output part. At the input part, power supply will supply the power to the microcontroller so that can operate the microcontroller. The parameter that will be measured including temperature, humidity, smoke gas around the hives and weight of the hives are measured by using suitable sensors.

The data that have been measured will be sent to the microcontroller to be process. Microcontroller will execute the programs from Energia software for controlling the sensor device. The microcontroller then will transmit the data to be displayed at memory LCD display.

IoT will connect the embedded system to the Cloud meaning the data measured will be available on the internet. Wi-Fi gives Internet connection to devices where in the range of a wireless network that have Internet connection. MQTT protocol will subscribe and publish the data in the Cloud. MQTT lens publisher and subscriber test tool were used, and add-on to Chrome to confirm the availability of the data in the cloud parked at a free cloud server. Node Red will be used for designing the User Interface for viewing the data so that the data can be displayed on graph, chart, gauge etc. This Node-RED will reside on a server, also in the Cloud- Cloud server. It will be setup later. Temporary, free cloud server is used at iot.eclipse.org. User can monitor the data at the webpage on any smart devices or anything else.

E. Conceptual Design

The conceptual design is shown in Fig. 4. The 4 load 20 kg sensors are placed at each side corner to construct the scale that can carry maximum weight 80 kg of beehives weight can be monitored as shown in Fig. 5.

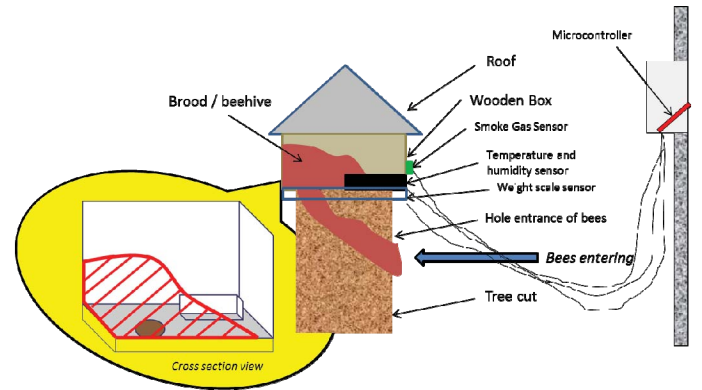


Fig. 4. Conceptual design of the system

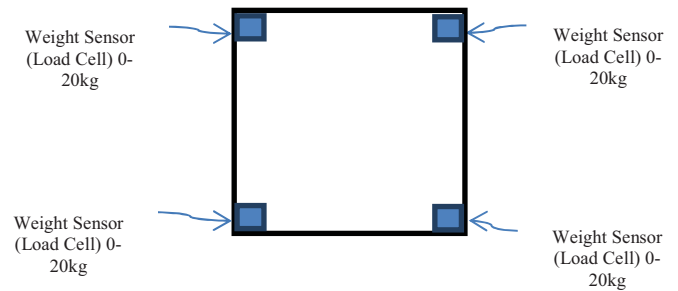


Fig. 5. Position of weight sensors

III. RESULTS AND DISCUSSIONS

Figure 6 shows the graph of weight scales reading over the real-time. From the graph, shown that the weight decreases at evening until at about (18:00:00) and increasing until (19:29:00), then the data constant until at (20:10:00). The data were decreasing because it was raining at that time, so the bees going out of the hives. It was increasing back because the hives were getting wet because of raindrops and increase humidity on the hives.



Fig. 6. Weigh scale data

Figure 7 shows the list of current value where the real-time value at the time data was captured at (20:10:00). The humidity was high and the temperature was optimum. The humidity was high because it was raining at that time. The hive was normal and in good condition.

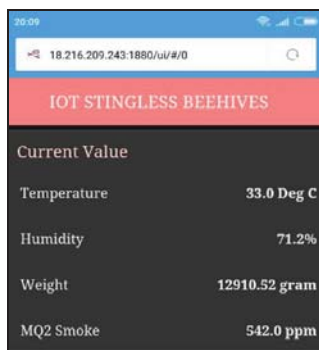


Fig. 7. List of current value

Figure 8 shows that the temperature inside the hive was kept constant at all time, referring to both graphs of Temperature over real-time and Humidity over real-time, there was not a big change at all time at about 33.0°C till 33.90°C. While humidity data were constant around 70% till 75%.

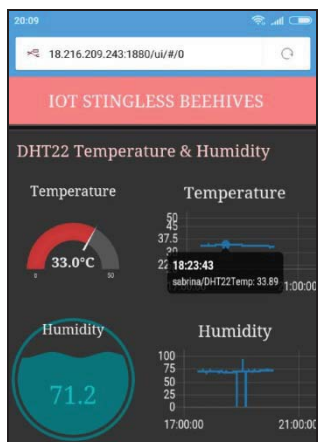


Fig. 8. Temperature variations

Referring to the figure 9, the gauge shows the MQ2 smoke sensor data. This data should display in ppm unit, but since the formula is quite complicated, it only displays in analog data. That shows the normal analog data (545) for the sensor at clean air which there was no smoke around the hive.

Other than that, the weight sensor was functioning well but the value was not accurate, as when there was no load, sometimes the reading shows a negative value. It was hard to calibrate it well and because of the HX711 ADC 24-bit converter itself. There were caused by many parameters that important to be considered. One of them was the sensitivity (rated output) of the load cells defines as full load output voltage in relation to output voltage where high precision ADC mandatory when dealing with load cells. Load cells also have own non-linearity based on their construction. Hysteresis error is the change in load cell output when a known load is reached from a lesser weight as compared to when it is reached from a higher weight. Other, repeatability was the change in the load measured by the load cell when the same weight is placed multiple times on the same load cell. Creep and creep recovery also important means measurements of change in measured weight over time when a weight was placed on a scale for a long period of time. When the weight is just placed and 30 minutes after, the output counts will be different. This effect is based on the elastic property of the material used in the load cell where it may take a short or long time for the load cell to recover from the deformation.

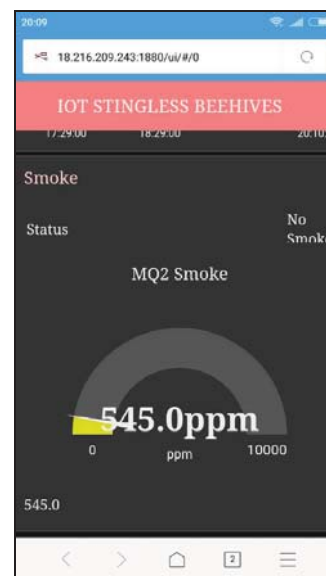


Fig. 9. Smoke sensor data

From the reading data captured between times at (17:29:00) until (20:10:00), the data shows the stingless bees behaviour where they will leave the hive when it was raining shown from weight data. The temperature 33.0°C till 33.90°C and humidity 70% till 75% remain constants at the optimum readings. This show the stingless bees were in good health since the bees will die especially to the pupae if the

temperature reaches 38°C. The smoke sensor reading shows the good air quality for stingless bees because stingless bees were very sensitive to air surroundings. [17]

IV. CONCLUSION

This paper described on the investigation of parameters that affect the behaviour of the beehives and production of honey. The sensors selected are DHT22 to measure temperature and humidity inside the beehives, the MQ2 smoke gas sensor to monitor the smoke surrounding the beehives, and weight scale to measure the weight of the beehives in real time. The sensors are then integrated and tested on the CC3200 microcontroller. Internet of things platform are integrated and investigated by monitoring the real-time data on the User Interface, Node-RED on the computer, tablet, and phones through the Internet. The prototype for monitoring stingless beehives and bee health is then demonstrated and test to get the data on the beehives.

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